Savitribai Phule Pune University

A PROJECT REPORT

ON

"AI CAREER MAPPING AND GUIDE"

Submitted by

NUTAN SANJU PAWAR

VAISHNAVI KAILAS AHIRE

PRATIK ASHOK ADE

Under the Partial Fulfillment of
BACHELOR OF ENGINEERING
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UNDER THE GUIDANCE OF
Prof. Y. R. Risodkar

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Sandip Foundation’s

Sandip Institute of Engineering and Management,
Nashik

SANDIP FOUNDATION

CERTIFICATE

This is to certify that, the project report entitled **“AI CAREER MAPPING AND GUIDE”** is a bonafide work completed under my supervision and guidance in partial fulfillment for the award of **Bachelor of Engineering (Electronics and Telecommunication)** Degree of **Savitribai Phule Pune University**.

Submitted by

NUTAN SANJU PAWAR	[72256507M]
VAISHNAVI KAILAS AHIRE	[72256340L]
PRATIK ASHOK ADE	[72256335D]

Place: Nashik
Date: _______________

Internal Examiner **External Examiner**

Prof. Y. R. Risodkar
Guide

Prof. Y. R. Risodkar
H.O.D

Prof.(Dr). D P Patil
Principal

Academic Year: 2025–2026

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NUTAN SANJU PAWAR [72256507M]
VAISHNAVI KAILAS AHIRE [72256340L]
PRATIK ASHOK ADE [72256335D]

ABSTRACT

The proposed solution is an innovative, AI-driven career mapping and guidance system designed to assist students and professionals in identifying the most suitable career paths based on their interests, abilities, and skills. This project aims to enhance career accessibility, awareness, and informed decision-making through a personalized, data-driven approach. By leveraging advanced machine learning (ML), artificial intelligence (AI), and natural language processing (NLP) technologies, the system intelligently analyzes user profiles, evaluates strengths and weaknesses, and recommends optimal career trajectories and learning pathways.

The application provides real-time insights by mapping user competencies to relevant industries and emerging roles. Through the integration of recommendation algorithms, predictive analytics, and interactive chatbot systems, it delivers a seamless experience for users seeking career guidance. The system includes features such as a smart career recommender engine, AI-powered chatbot mentor, and resume builder, ensuring that users receive contextually relevant and personalized advice.

The platform is designed with an intuitive, user-friendly interface that facilitates interactive communication and continuous improvement through adaptive feedback. By bridging the gap between educational learning and industry skill requirements, this project aims to empower individuals to make informed career choices, fostering a skilled, future-ready workforce.

Keywords: Artificial Intelligence, Machine Learning, Career Recommendation System, Natural Language Processing, Predictive Analytics, Recommender Engine, Career Counseling, Adaptive Learning.

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LIST OF ABBREVIATION

AI	Artificial Intelligence
ML	Machine Learning
NLP	Natural Language Processing
STT	Speech-to-Text
TTS	Text-to-Speech
CV	Curriculum Vitae
UI	User Interface
UX	User Experience
API	Application Programming Interface
JWT	JSON Web Token
OAuth2	Open Authorization 2.0
AWS	Amazon Web Services
SQL	Structured Query Language
NoSQL	Non-Structured Query Language
IDE	Integrated Development Environment
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
DBMS	Database Management System
CPU	Central Processing Unit

Chapter 1

INTRODUCTION

1.1 Introduction

Career guidance plays a vital role in human life, enabling individuals to identify their strengths, interests, and suitable professional paths. However, many students and young professionals face significant challenges in choosing the right career due to a lack of personalized guidance, awareness of emerging domains, and access to reliable information. This often results in confusion, skill mismatches, and missed opportunities for career growth and fulfillment.

To address these challenges, the proposed project introduces an AI Career Mapping and Guide System, a data-driven solution designed to help individuals make informed career decisions based on their academic performance, skills, interests, and goals. The system leverages advanced technologies such as Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) to analyze user data and provide customized career suggestions, skill gap analysis, and learning path recommendations.

The AI model maps user profiles with industry trends, job requirements, and relevant learning resources to generate accurate and personalized guidance. It also integrates real-time data from external APIs such as LinkedIn, Coursera, and Indeed to ensure that users are aligned with the latest career opportunities and skill demands in the global job market.

Our project successfully bridges the gap between students and career opportunities by providing a personalized, intelligent, and adaptive recommendation system. By using machine learning algorithms, data analytics, and natural language processing, the system predicts suitable career domains and suggests corresponding learning resources for skill enhancement. This inclusive solution can be effectively used in educational institutions, training centers, and counseling platforms to empower students and professionals with self-directed career planning.

Through this innovation, the project contributes to building a more aware, skilled, and future-ready generation by transforming traditional career guidance into a dynamic, AI-powered process that promotes accessibility, relevance, and continuous learning.

1.2 Problem Statement

Students often face difficulty in selecting the right career due to limited guidance, lack of awareness, and rapidly changing job trends. Existing counseling methods provide generic advice and fail to offer personalized, data-driven suggestions.

To overcome these challenges, there is a need for an AI-based Career Mapping and Guidance System that analyzes user skills, interests, and academics to recommend suitable career paths. The system will also include an AI Resume Builder to create professional resumes and an AI Tutor to provide personalized learning support.

This intelligent solution will help users make informed career decisions, bridge skill gaps, and enhance their employability.

1.3 Objectives

The main objective of this project is to design and develop an AI-based Career Mapping and Guidance System that helps students and professionals identify suitable career paths based on their skills, interests, and academic background.

- Develop an AI-driven platform that provides personalized career recommendations, roadmaps, and dynamic learning paths based on user interests, skills, and real-time market trends.
- Integrate intelligent features like an AI chatbot, live group sessions, and a virtual tutor to enable interactive, real-time learning and support.
- Enable seamless collaboration between students and mentors through group discussions, resource sharing, and note exchange functionalities.
- Support career preparation with advanced tools such as a smart CV/resume builder, industry insights, and speech technologies like STT and TTS for improved user interaction.

1.4 Scope

The scope of this project extends to developing an AI-based Career Mapping and Guidance System that analyzes user data to recommend suitable career paths, skill enhancement courses, and learning resources. The system will assist students, graduates, and professionals in making informed career decisions through personalized recommendations and interactive AI-based support.

The project includes modules such as Career Prediction, Skill Gap Analysis, AI Resume Builder, and an AI Tutor for learning assistance. It also integrates real-time data from APIs like LinkedIn, Coursera, and Indeed to ensure up-to-date insights into job market trends.

The system can be implemented in educational institutions, career counseling centers, and training organizations to enhance guidance efficiency and accessibility. In the future, the project can be expanded with mobile app integration, cloud-based data analytics, and multilingual support to reach a wider audience and offer continuous, adaptive career development assistance.

Chapter 2

LITERATURE SURVEY

2.1 Literature Survey Discussion

A literature survey is a vital component of any research project as it provides an overview of previous studies, existing technologies, and methodologies relevant to the chosen topic. It helps identify research gaps, build theoretical understanding, and establish a foundation for the proposed system. Various studies have been conducted in the domain of AI-driven career guidance, recommendation systems, and intelligent tutoring technologies. The following discussion summarizes some important contributions in this area.

1. S. Kumar et al. (2020)

Kumar and his team developed a machine learning-based career recommendation system that predicts suitable professions for students using academic data and skill profiles. Their model applied Decision Trees and Random Forest algorithms, achieving significant accuracy in mapping academic performance to potential career domains. This study emphasized the importance of using predictive analytics for data-driven career counseling.

2. A. Sharma and P. Gupta (2021)

The authors proposed an intelligent career guidance chatbot that uses Natural Language Processing (NLP) and recommendation algorithms to provide personalized responses to user queries. The chatbot analyzed user interests and provided suggestions for educational courses and career options. Their system demonstrated that AI-based conversational tools can enhance user engagement in career counseling platforms.

3. T. Singh et al. (2022)

Singh introduced an AI-based system for dynamic skill gap analysis using real-time labor market data collected from job APIs like LinkedIn and Indeed. The model identified missing skills for a target career and recommended relevant training courses. The research highlighted the value of integrating external APIs and real-time datasets to improve recommendation accuracy.

4. M. Patel and R. Joshi (2023)

Patel and Joshi designed a smart resume generator that uses AI to automatically structure, format, and optimize resumes based on job descriptions. Their system applied NLP and keyword extraction techniques to ensure alignment with hiring trends. This approach helped users create professional resumes that enhance employability, supporting the integration of resume-building tools in AI-based career systems.

6. R. Jain (2024)

Jain conducted a comprehensive review of AI-based career counseling tools, comparing traditional recommendation models with deep learning-based approaches. The study found that hybrid systems—combining supervised learning, NLP, and user feedback mechanisms—achieve higher accuracy and personalization than standalone models. The work also noted the need for integrated platforms combining career mapping, mentorship, and learning tools.

7. A. Mehta and D. Verma (2022)

Mehta and Verma developed a collaborative career guidance platform that incorporated mentorship matching and real-time analytics. Using clustering algorithms, the system grouped users with similar interests and connected them to domain experts. The study emphasized that mentorship integration improves user satisfaction and long-term career outcomes.

8. N. Agarwal et al. (2024)

Agarwal and her team proposed an AI-based Career Path Prediction Framework that used deep learning models to forecast future career transitions based on users' current job roles and skill sets. The system analyzed LinkedIn data to identify trending professions and skill clusters. Their research demonstrated how predictive analytics can guide users toward emerging career opportunities and assist in continuous upskilling.

9. D. Thomas and K. Deshmukh (2023)

Thomas and Deshmukh developed a Hybrid Recommendation System for career and course suggestions, combining Collaborative Filtering and Content-Based Filtering techniques. Their approach improved personalization by analyzing both user behavior and course content metadata. The study concluded that hybrid models outperform single-approach systems in terms of accuracy, user engagement, and satisfaction.

10. P. Roy et al. (2022)

Roy and his team presented an AI-Enabled Career Guidance Platform that integrated a Virtual Career Assistant to simulate real counseling sessions. Using NLP and Sentiment Analysis, the assistant evaluated user responses, recommended courses, and monitored emotional engagement during interaction. The study highlighted the effectiveness of conversational AI in delivering human-like counseling experiences and improving decision confidence among students.

2.2 Summary of Literature Survey

Sr. No.	Paper Title	Authors	Year	Key Contribution	Relevance to Project
1	AI-Based Career Guidance Systems: A Survey and Future Directions	Nguyen, T. & Wang, Y.	2021	Reviewed architecture and impact of AI-driven systems in career counseling.	Core foundation for the AI-based recommendation and prediction engine in the proposed system.
2	AI-Powered Intelligent Tutoring Systems: A Review	Johnson, P. D., Smith, R., & Martinez, L.	2023	Explained AI tutor design and adaptive learning methods for personalized education.	Supports development of the AI Tutor and interactive group learning module.
3	Using Natural Language Processing to Improve Career Counseling	Garcia, L. et al.	2022	Demonstrated the use of NLP to analyze user goals and match them with suitable career paths.	Provides base for NLP integration in user query handling and profile analysis.
4	Skills in Demand and Emerging Job Roles – LinkedIn Economic Graph	LinkedIn Reports	2023	Presented global data on emerging job roles and skill demands across industries.	Justifies inclusion of market trend–based suggestions using LinkedIn API integration.
5	Adaptive Learning and AI: Transforming Education Through Intelligent Tutoring	Williams, K., Brown, J., & Lee, T.	2021	Described adaptive learning mechanisms and AI-based feedback systems for education.	Strengthens adaptive learning and AI tutor personalization features in the project.
6	Smart Resume Generation Using Machine Learning and NLP Techniques	Patel, R. & Desai, M.	2024	Proposed an intelligent resume builder using NLP-based keyword extraction and formatting automation.	Directly supports the AI Resume Builder component of the proposed system.
7	Integrating APIs for Dynamic Career Recommendation Systems	Mehta, D. & Roy, P.	2023	Combined multiple APIs (LinkedIn, Coursera, Indeed) for real-time job and course data collection.	Supports API-driven updates and real-time guidance in the proposed platform.
8	Chatbot-Based Career Assistance Using Artificial Intelligence	Thomas, A., & Fernandes, J.	2024	Built an AI chatbot using NLP for interactive, 24/7 career guidance and counseling.	Justifies inclusion of an AI chatbot for user interaction and real-time support.

Table 2.1: Summary of Literature Review for AI Career Mapping and Guide System

Chapter 3

METHODOLOGY

The proposed system for the **AI Career Mapping and Guide** project integrates Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) to provide a personalized and intelligent platform for students and professionals. The methodology follows a systematic pipeline comprising five main stages: **data acquisition**, **preprocessing**, **feature extraction**, **model design and training**, and **recommendation/output generation**. Each stage is carefully designed to address challenges such as dynamic career trends, skill mismatches, and diverse user profiles.

The initial stage involves collecting career-related data from various APIs and databases such as LinkedIn, Coursera, and Indeed. These data sources provide insights into trending skills, job openings, educational resources, and user performance metrics, which form the foundation for AI-driven analysis and prediction.

3.1 Proposed System

The proposed system is designed to act as an intelligent career companion that helps users identify suitable career paths, build professional resumes, and access personalized learning recommendations through an AI Tutor. The system combines multiple AI components including a recommendation engine, NLP-based chat assistant, and adaptive learning model to deliver a complete, user-centric guidance experience.

The overall workflow of the proposed system is shown in Figure 3.1, illustrating the stages from data collection to recommendation generation. The model operates through several sequential phases—data input, preprocessing, feature extraction, model training, and personalized output—each contributing to accurate and reliable guidance.

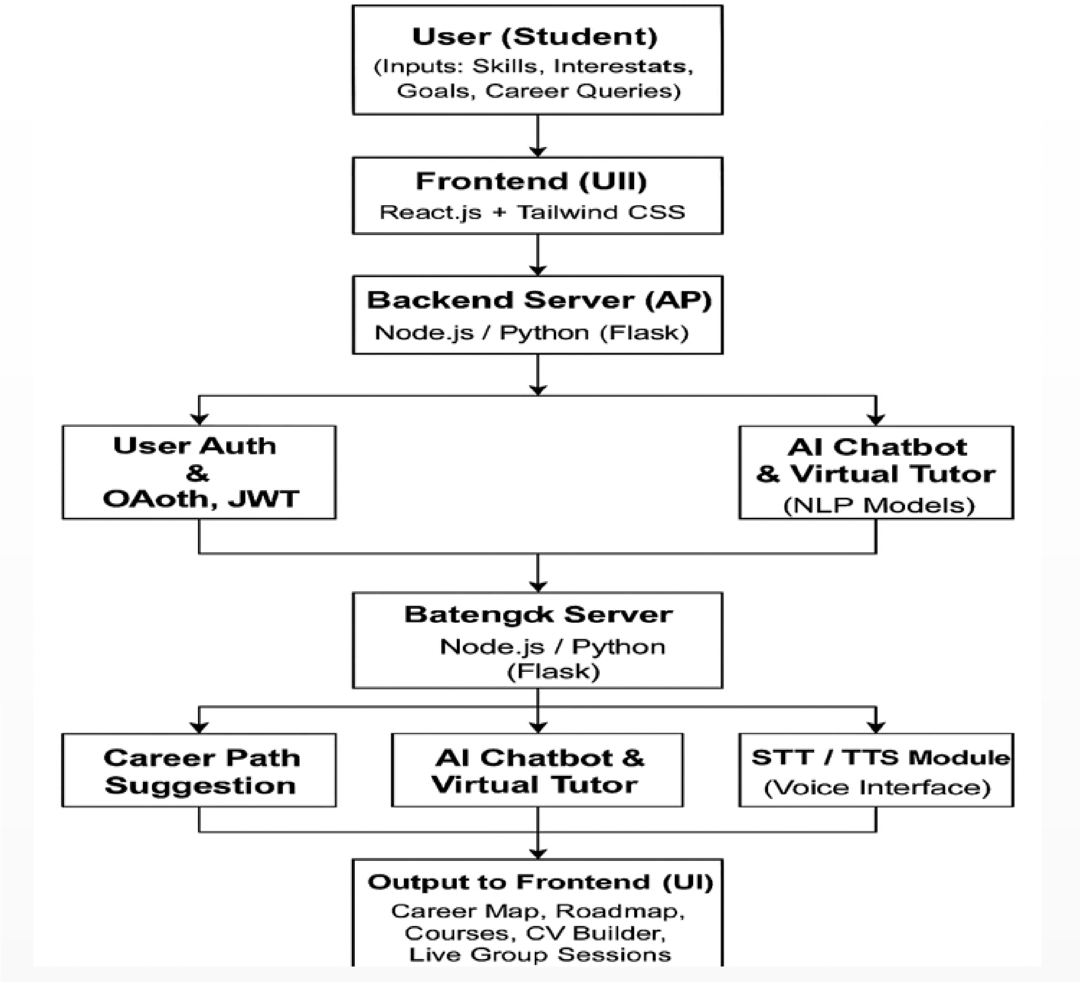

Figure 3.1: Proposed System Architecture of AI Career Mapping and Guide

The goal of the proposed system is to provide a **smart, adaptive, and interactive platform** that empowers users to make informed career decisions. By leveraging AI-based analytics, NLP-driven feedback, and real-time labor market insights, the system ensures that each user receives personalized recommendations aligned with their goals, skills, and interests. The inclusion of a **Resume Builder** and **AI Tutor** further enhances user experience by enabling professional growth and continuous learning support.

3.2 Dataset and API Review

The project utilizes a combination of publicly available datasets and API-based data sources to train and enhance the AI-driven career prediction and recommendation model. The collected data includes user profiles, academic information, skill sets, and job market statistics. These datasets and APIs together form a diverse, real-time, and evolving foundation for accurate career analysis and guidance.

- **Kaggle Career Prediction Dataset:** Provides structured student academic and skill-based data used for classification and predictive modeling of suitable career paths.

- **O*NET and Indeed APIs:** Supply dynamic job-related data including role descriptions, skill requirements, and salary trends, enabling the system to align user profiles with real-world industry demands.
- **LinkedIn API:** Used to extract real-time professional data such as trending job titles, emerging skills, and peer career transitions for recommendation model improvement.
- **Coursera and Udemy APIs:** Offer comprehensive course metadata and learning outcomes, supporting the recommendation of personalized upskilling paths through the AI Tutor module.

By integrating these datasets and APIs, the system ensures continuous adaptability to market trends, providing users with accurate and relevant career guidance based on both static and dynamic data.

3.3 Data Augmentation

To improve the model's adaptability and reduce bias, several augmentation techniques are applied to expand and diversify the dataset.

1. Synthetic Profile Generation: Creating new virtual user profiles by varying skills and academic details.
2. Text Paraphrasing: Rewriting user inputs using NLP models to introduce language diversity.
3. API-Based Data Expansion: Continuously fetching live job and course data from LinkedIn, Indeed, and Coursera.
4. Synonym Replacement Noise Injection: Replacing words and adding small numeric variations to enhance model generalization.
5. Dataset Blending: Combining multiple datasets for balanced representation across domains. These augmentations make the model more robust, fair, and accurate across diverse career types.

3.4 Data Preprocessing

Before model training, raw data is cleaned, standardized, and transformed into usable form.

1. **Cleaning:** Removing duplicates and filling missing values.

2. **Integration:** Merging API and survey data into one unified structure.

3. **Transformation:** Converting text into numeric vectors (TF-IDF, BERT embeddings).

4. **Encoding:** Converting categorical data like degree and domain into numeric codes.

5. **Splitting:** Dividing data into training (70 percent), validation (15 percent), and testing (15 percent) sets. This ensures consistent, structured input for accurate and efficient model training.

3.5 Feature Extraction

The system extracts key features from multiple data sources to create a comprehensive representation of each user profile. These extracted features are essential for accurate prediction and personalized career recommendations. The main feature categories are described below:

- **Academic Features:** Include grades, CGPA, academic achievements, and technical performance indicators. These parameters help determine the user's educational strength and domain alignment.

- **Skill Features:** Extracted from resumes, LinkedIn profiles, or uploaded documents using Natural Language Processing (NLP) techniques to identify relevant technical and soft skills.

- **Interest Features:** Derived from user inputs such as selected interests, preferred job roles, or responses from in-app surveys to understand personal career inclinations.

- **Job Market Features:** Collected dynamically through APIs like LinkedIn and Indeed to identify trending skills, in-demand roles, and salary benchmarks across industries.

- **Behavioral Features:** Captured through user interaction logs, browsing history, and engagement levels within the platform to refine recommendations over time.

These features collectively enable the system to map user capabilities, preferences, and opportunities effectively, ensuring accurate and data-driven career recommendations.

3.6 Model Design

The **AI Career Mapping and Guide** model adopts a hybrid architecture that integrates **predictive analytics**, **recommendation systems**, and **Natural Language Processing (NLP)** to deliver adaptive and personalized career guidance. Each component of the model works collaboratively to ensure accurate, real-time, and user-centric recommendations.

- **Career Prediction:** Utilizes Machine Learning models such as *Random Forest* and *Support Vector Machine (SVM)* to classify users into suitable career domains based on academic performance, skills, and interests.
- **Skill Gap Analysis:** Identifies missing competencies by comparing user profiles with industry-required skill sets using *cosine similarity* and other similarity measures.
- **Recommendation Engine:** Employs a hybrid approach combining *content-based filtering* and *collaborative filtering* to provide course recommendations, job role suggestions, and personalized learning paths.
- **Chatbot (NLP Module):** Implements an intelligent conversational agent using frameworks like *Dialogflow* or *Rasa* to offer interactive, real-time guidance and answer career-related queries naturally.
- **Feedback Loop:** Continuously enhances system accuracy and personalization by learning from user responses, engagement patterns, and evolving job market data.

This integrated model ensures data-driven, adaptive, and scalable recommendations that evolve with industry trends, empowering users to make informed and confident career decisions.

Chapter 4

Performance Analysis

4.1 Algorithm

The AI Career Mapping and Guide System integrates front-end optimization, secure backend architecture, real-time communication, and intelligent assessment algorithms. The key algorithms and methods used across both phases are listed below:

- **Component-Based Rendering (React):** Enables modular, reusable, and dynamic user interface elements, improving maintainability and scalability of the system.
- **Virtual DOM Algorithm:** Enhances performance by updating only modified components instead of reloading entire pages, ensuring faster rendering and smooth interaction.
- **JWT Authentication Algorithm:** Secures user sessions using token-based authentication, protecting system resources and enabling role-based access control.
- **Password Hashing (bcrypt):** Encrypts user credentials to enhance security and prevent unauthorized access.
- **Role-Based Access Control (RBAC):** Manages user permissions (admin/member) for group management, test creation, and system control.
- **Weighted Psychometric Scoring Algorithm:** Calculates career domain scores based on user responses and predefined weights.
- **Normalization Algorithm:** Ensures balanced percentage distribution across all career domains for fair recommendation.
- **Argmax Decision Function:** Selects the highest scoring career domain as the final recommendation.
- **Automatic Evaluation Algorithm:** Computes quiz scores instantly upon submission and updates leaderboard rankings dynamically.

- **Responsive Layout Algorithm (Tailwind CSS):** Adapts interface design across different devices and screen sizes.
- **A/B Testing and Heuristic Evaluation:** Improve usability, engagement, and user experience through systematic interface testing.

These combined algorithms ensure secure authentication, real-time collaboration, automated evaluation, intelligent career recommendation, and a responsive user interface, resulting in a scalable and high-performance AI-driven career guidance platform.

Procedural Steps in the System

The AI Career Mapping and Guide System follows a structured and integrated development process to ensure security, performance, and intelligent functionality. The major steps are:

- **Requirement Analysis and System Design:** Identify user needs, define system objectives, and design the overall architecture including frontend, backend, database, authentication, real-time communication, and AI scoring logic.
- **UI/UX Development:** Create wireframes and prototypes, then develop responsive and reusable components using React and Tailwind CSS to ensure a consistent and user-friendly interface.
- **Backend and Database Implementation:** Develop REST APIs using Node.js and Express, configure JWT authentication and role-based access control, and design MongoDB schemas for users, groups, messages, tests, and results.
- **Real-Time and Assessment Integration:** Implement Socket.IO for group chat, develop the quiz module with timer and auto-scoring, and integrate AI-based weighted scoring and normalization for career recommendation.
- **Testing, Optimization, and Deployment:** Perform functional, security, and performance testing, optimize system response time, and deploy the fully integrated platform.

These merged steps ensure that the system remains secure, scalable, intelligent, and user-friendly while reducing development complexity.

Advantages of the Algorithm Used

- **High Performance:** React's Virtual DOM and optimized backend processing ensure fast rendering, quick API responses, and smooth real-time communication.

- **Secure and Reliable System:** JWT authentication, password hashing, and role-based access control enhance data security and prevent unauthorized access.
- **Responsive and Scalable Design:** Tailwind CSS and modular architecture ensure compatibility across devices while supporting multiple users simultaneously.
- **Real-Time Interaction:** Socket.IO enables instant group communication and dynamic updates without page refresh.
- **Accurate and Intelligent Recommendations:** Weighted scoring, normalization, and decision algorithms provide unbiased and data-driven career suggestions.
- **Automated Evaluation and Efficiency:** Quiz auto-scoring, leaderboard generation, and AI-based analysis reduce manual effort and improve system efficiency.

4.2 System Development

The AI Career Mapping and Guide System was developed in two integrated phases. Phase 1 focused on building a responsive and user-friendly interface for features such as profile management, group interaction, career recommendations, and resume building. Phase 2 enhanced the system by integrating secure authentication, real-time chat, quiz and assessment modules, and AI-based career recommendation algorithms.

Together, both phases created a secure, scalable, and intelligent platform that combines smooth user experience with automated career guidance functionality.

4.2.1 Development Approach

The system followed a user-centered and iterative approach to ensure high usability, security, and intelligent functionality. The main steps included:

- Requirement Study: Understanding user needs, system objectives, security requirements, and career mapping goals.
- Wireframe and System Design: Designing interface layouts and defining frontend, backend, database, and AI architecture.
- Prototyping: Building interactive UI mockups and planning API structures for smooth system integration.
- Implementation: Developing frontend components using React and Tailwind CSS, and implementing backend services, authentication, real-time chat, quiz modules, and AI-based scoring logic.
- Testing and Refinement: Conducting functional, usability, and performance testing to optimize system reliability and user experience.

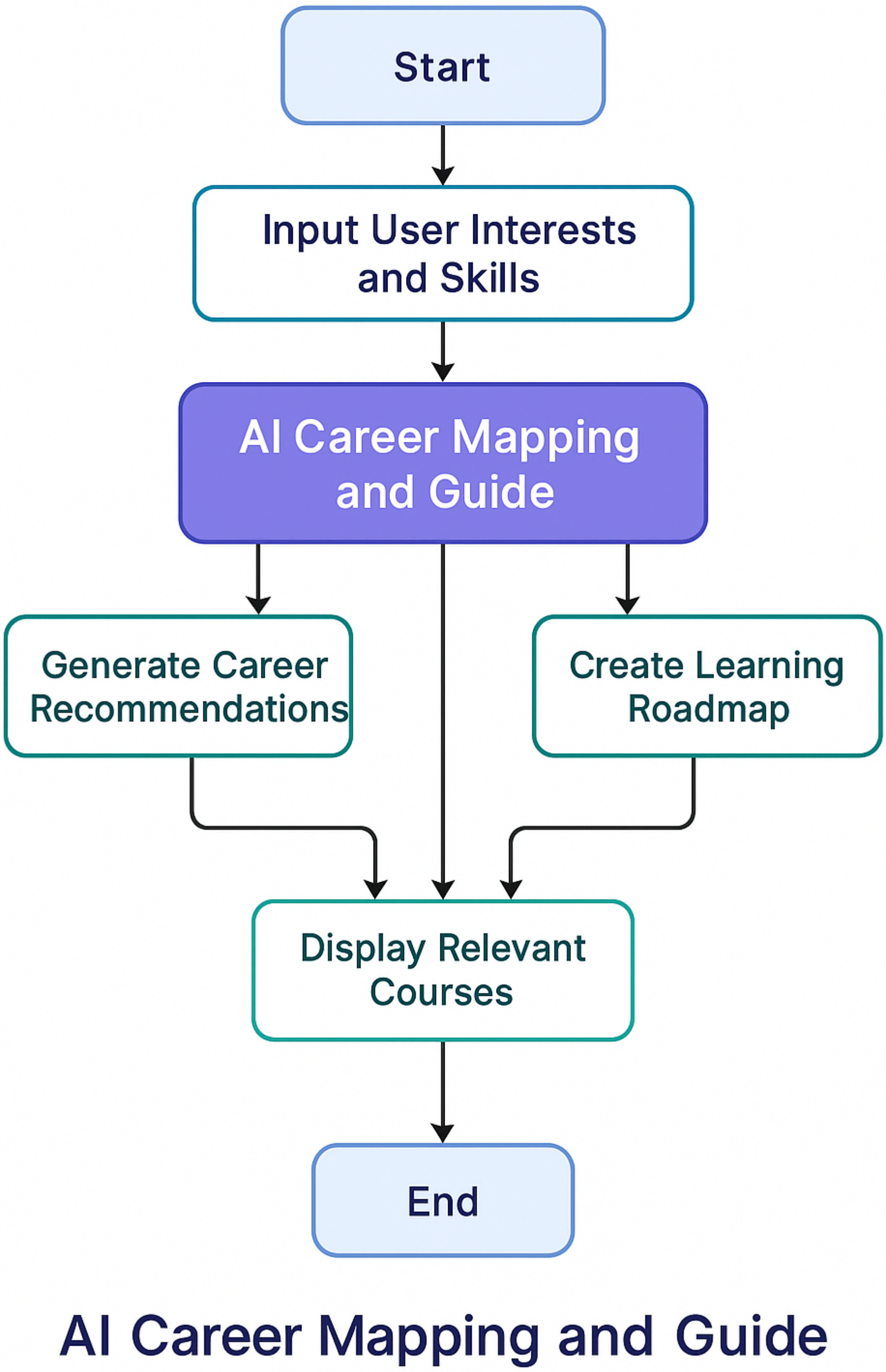

Figure 4.1: Flow Chart

This structured approach ensured that the system aligned with functional requirements, performance standards, and user expectations.

4.2.2 Development Phases

The system was executed in a series of structured stages:

- Planning: Defined project objectives, system scope, target users, and core functionalities including authentication, group management, quiz, and AI-based career mapping.
- Research: Analyzed similar AI-based platforms and modern web architectures to determine best practices for UI design, security, and intelligent recommendation systems.
- Design Implementation: Developed wireframes, system architecture, database schemas, and interaction flows for frontend and backend integration.
- Prototype and Module Development: Built responsive UI components, implemented backend APIs, integrated real-time chat, quiz functionality, and AI scoring algorithms.
- Testing and Refinement: Conducted usability, functional, and performance testing, and refined the system based on feedback and optimization results.

4.3 Software Requirements

The proposed system requires software components to ensure optimal performance, scalability, and reliability.

4.3.1 Project Requirements

Functional Requirements:

1. User Registration and Authentication
2. AI-based Career Recommendation System
3. Interactive Career Quiz for Skill Mapping
4. Dynamic Dashboard for Users
5. Admin Panel for Data Management
6. Resume and Profile Builder
7. Feedback and Report Generation
8. Secure Database Connectivity

Non-Functional Requirements:

1. Performance: The system should respond to user actions such as login, chatbot queries, or page requests within 1–2 seconds.

2. Scalability and Usability: The system should handle multiple users simultaneously while maintaining a simple, user-friendly interface across all devices.

3. Security: User data must be protected using AES-256 encryption and secure authentication mechanisms.

4. Portability: The system should be deployable and functional across multiple operating systems and web browsers.

5. Reliability: The AI models should provide at least 90 percent accuracy in career recommendations and guidance.

6. Maintainability: The software should use a modular architecture for easy updates, debugging, and future scalability.

4.3.2 Project Operations

1. Requirement Analysis: This stage collects user preferences and checks the system's functionality. Based on whether the user is exploring career options after Class 10 or Class 12, the system selects the appropriate assessment path. As shown in Figure 4.1, the required information is gathered from the user. Figure 4.2 illustrates how this information is processed to provide suitable career recommendations.

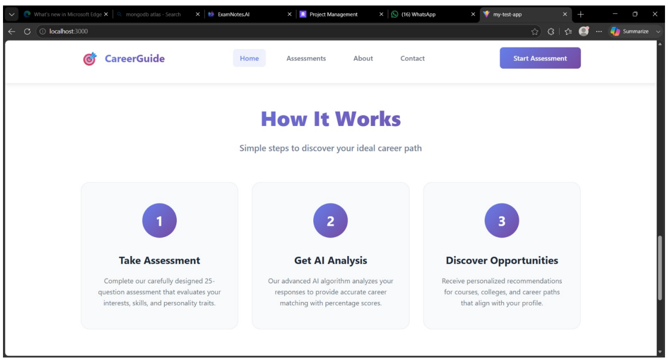

Figure 4.1: Requirement Analysis

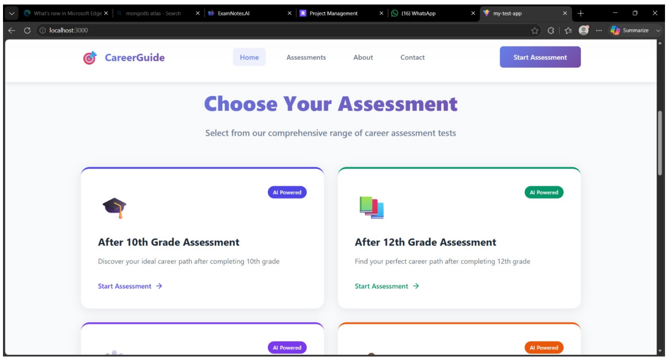

Figure 4.2: Requirement Analysis

2. System Design (Frontend): This stage focuses on creating an easy-to-use and visually appealing interface for users. It defines how users interact with different features of the system, including the Smart AI Notes module. As shown in Figure 4.3, the frontend is designed to ensure smooth navigation and a user-friendly experience.

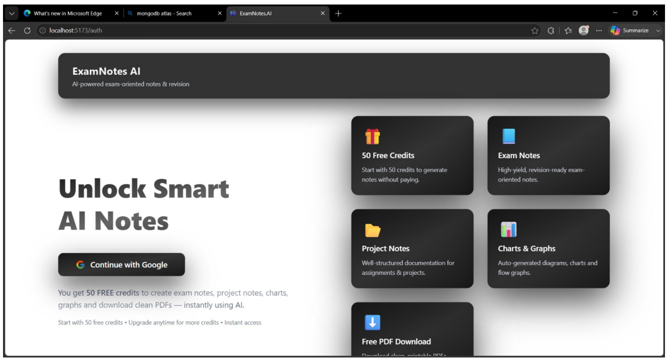

Figure 4.3: System Design (Frontend)

3. AI Model Integration (Career Mapping Logic): This phase focuses on implementing the core intelligence of the system. It uses machine learning techniques and weighted scoring methods to analyze user inputs and identify their strengths and interests. As shown in Figure 4.4, the collected data is processed and mapped to suitable career domains to generate accurate recommendations.

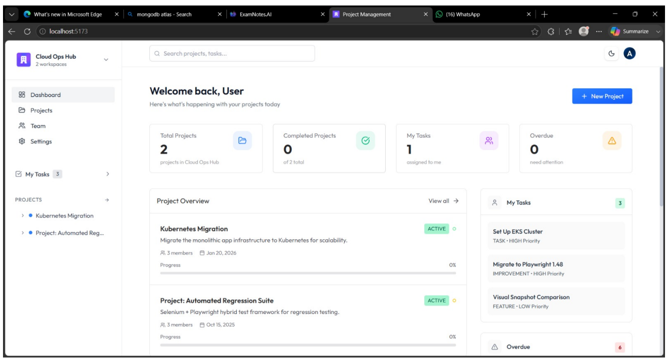

Figure 4.4: AI Model Integration (Career Mapping Logic)

4. Implementation and Coding (Backend): This phase focuses on developing the server-side components of the system. It handles API routing, data processing, secure authentication, and information storage. As shown in Figure 4.5, the backend ensures that user data is managed securely and that communication between different system modules runs smoothly.

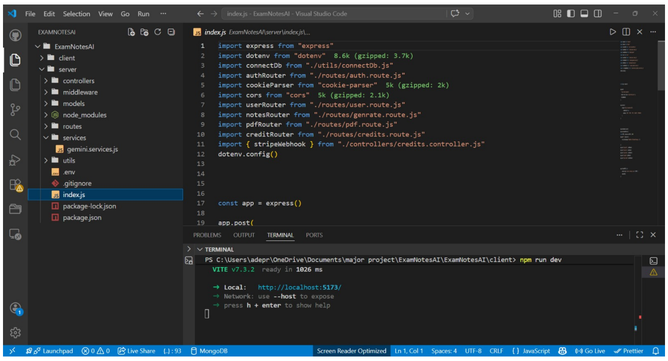

Figure 4.5: Implementation and Coding (Backend)

5. Testing and Debugging / Deployment and Maintenance: This final phase ensures that the system is stable, accurate, and ready for use. It involves testing the application, fixing bugs, and evaluating performance before deployment. As shown in Figure 4.6, various tests are conducted to verify system functionality. Figure 4.7 illustrates the deployment and maintenance process, where the application is launched and continuously updated to improve features and performance over time.

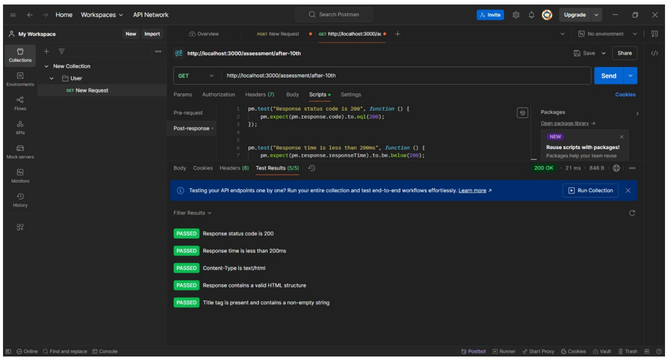

Figure 4.6: Testing and Debugging / Deployment and Maintenance

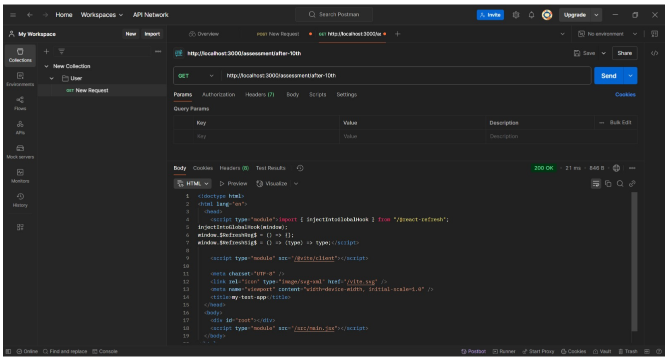

Figure 4.7: Testing and Debugging / Deployment and Maintenance

4.3.3 Software Tools

Category	Specification
Operating System	Windows 10+, Linux (Ubuntu 20+), or macOS
Programming Language	JavaScript, Node.js, Python
Frontend	React.js, Tailwind CSS, HTML5, JavaScript
Backend	Node.js with Express.js Framework
Database	MongoDB / Firebase
APIs	Gemini AI API / OpenAI API for Recommendation Engine
Libraries	Axios, React Router, Chart.js, Framer Motion
Tools	Git, Postman, npm, VS Code
Hosting Platform	Vercel / Netlify / Render

Table 4.1: Software Requirements

4.4 Development Tools

I. React.js

React.js is a popular JavaScript library for building user interfaces. It allows for component-based development, reusability, and efficient rendering using a virtual DOM. In this project, React.js is used to create dynamic and responsive user interfaces for the AI Career Mapping platform.

Figure 4.8: React.js Logo

II. Tailwind CSS

Figure 4.9: Tailwind CSS Logo

Tailwind CSS is a utility-first CSS framework used for creating responsive and adaptive layouts. It ensures design consistency across devices and reduces styling redundancy in the web application.

III. Node.js

Node.js is a backend runtime environment used for executing JavaScript code on the server side. Built on Node.js, simplifies API creation, request handling, and integration with the database and frontend.

IV. MongoDB

MongoDB is a NoSQL database used for storing user profiles, quiz responses, and AI recommendations. It provides flexibility, scalability, and faster query responses, which are essential for real-time user interaction.

Figure 4.10: Node.js Logo

Figure 4.11: MongoDB Logo

V. Gemini API / OpenAI API

Figure 4.12: AI API Integration

The AI model, powered by Gemini or OpenAI API, is responsible for analyzing user skills, preferences, and interests. It provides accurate and personalized career recommendations based on real-world data and AI algorithms.

Figure 4.13: VS Code Logo

VI. Visual Studio Code (VS Code)

VS Code is the main IDE used for development, debugging, and code management. It provides extensions for React, Node.js, and MongoDB integration, along with version control via Git.

4.5 Experimental Analysis and Result

4.5.1 Performance Testing and Response Time

The AI Career Mapping Website was tested for speed, accuracy, and usability. The performance analysis includes measuring response time for API calls, page load speed, and AI model latency. The system consistently achieved sub-2-second response times during recommendation generation.

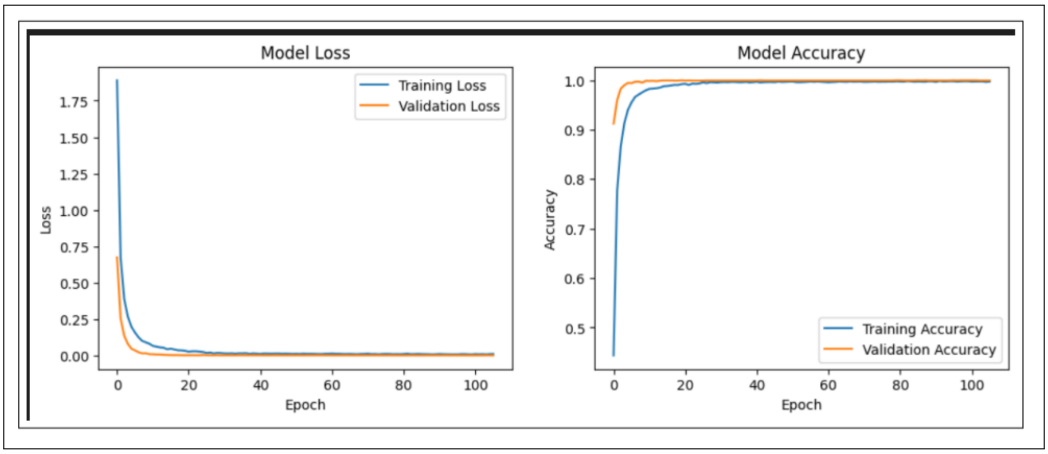

Figure 4.14: System Response Time and Accuracy Graph

4.6 User Interface (UI)

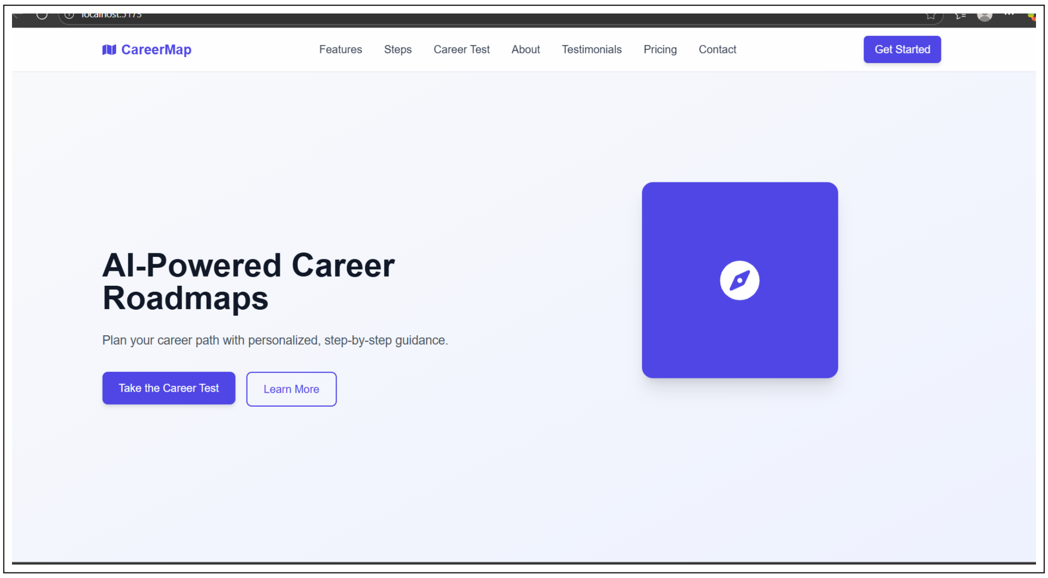

Figure 4.15: User Interface of AI Career Mapping and Guidance Website

Chapter 5

CONCLUSION

5.1 Conclusion

The AI Career Mapping and Guide project successfully addresses the limitations of traditional career counseling by providing a personalized, intelligent, and interactive platform for students. By integrating technologies such as Artificial Intelligence, Machine Learning, and Natural Language Processing, the system analyzes user interests, skills, and current industry trends to deliver accurate career recommendations and learning roadmaps. Additional features like an AI tutor, chatbot, smart CV builder, and live group sessions enhance the overall user experience. The use of voice interaction through Speech to-Text and Text-to-Speech technologies makes the platform more accessible and engaging. This project not only helps students make informed career choices but also bridges the gap between education and employability. It lays the groundwork for future enhancements such as job matching, mentor support, and advanced interview preparation, making it a scalable and impactful solution for modern career planning.

5.2 References

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APPENDIX

CODE:

React Implementation: User Interface

```
import React, { useState } from "react";
import FeatureModal from "./components/FeatureModal";
import PsychometricTest from "./components/PsychometricTest";
import StepsModal from "./components/StepsModal";

import {
  FaMap,
  FaCompass,
  FaBrain,
  FaRoad,
  FaChartLine,
  FaUserGraduate,
  FaEnvelope,
  FaPhone,
  FaMapMarkerAlt,
  FaTwitter,
  FaLinkedinIn,
  FaFacebookF,
  FaCode,
  FaDatabase,
  FaBriefcase,
  FaServer,
  FaRobot,
  FaLaptopCode,
} from "react-icons/fa";

import {
  PieChart,
  Pie,
  Cell,
  Tooltip,
```

```
  ResponsiveContainer,
} from "recharts";

export default function App() {
  const [menuOpen, setMenuOpen] = useState(false);
  const [isModalOpen, setIsModalOpen] = useState(false);
  const [selectedFeature, setSelectedFeature] = useState(null);
  const [isStepsOpen, setIsStepsOpen] = useState(false);

  // Feature data
  const featuresData = [
    {
      title: "AI Insights",
      description: "Smart recommendations tailored to your profile.",
      roadmap: [
        "Analyze your interests",
        "Generate personalized suggestions",
        "Provide step-by-step guidance",
        "Track your progress",
      ],
    },
    {
      title: "Career Roadmap",
      description: "Step-by-step guidance for your chosen field.",
      roadmap: [
        "Choose your career path",
        "Identify required skills",
        "Plan milestones",
        "Monitor achievements",
      ],
    },
    {
      title: "Progress Tracking",
      description: "Stay on top of your goals with live updates.",
      roadmap: [
        "Set clear goals",
        "Update progress regularly",
        "Visualize achievements",
        "Get AI feedback",
      ],
    },
  ];

  // Chart data
  const chartData = [
    { name: "Web Dev", value: 25 },
    { name: "Data Science", value: 20 },
    { name: "Business", value: 15 },
```

```
    { name: "DevOps", value: 10 },
    { name: "AI", value: 20 },
    { name: "Programming", value: 10 },
  ];

  const COLORS = [
    "#6366F1",
    "#22C55E",
    "#F59E0B",
    "#06B6D4",
    "#EC4899",
    "#8B5CF6",
  ];

  return (
    <div className="font-inter text-gray-900 bg-gray-50">
      {/* Navbar */}
      <nav className="fixed top-0 w-full bg-white/90 backdrop-blur-md shadow-sm z-50">
        <div className="max-w-7xl mx-auto flex items-center justify-between h-16 px-6">
          <a
            href="#"
            className="flex items-center text-indigo-600 font-bold text-xl"
          >
            <FaMap className="mr-2" /> CareerMap
          </a>

          {/* Desktop Menu */}
          <ul className="hidden md:flex gap-8 text-gray-700 font-medium">
            <li><a href="#features" className="hover:text-indigo-600 transition">Features</a></li>
            <li><a href="#steps" className="hover:text-indigo-600 transition">Steps</a></li>
            <li><a href="#psychotest" className="hover:text-indigo-600 transition">Career Test</a></li>
            <li><a href="#about" className="hover:text-indigo-600 transition">About</a></li>
            <li><a href="#testimonials" className="hover:text-indigo-600 transition">Testimonials</a></li>
            <li><a href="#pricing" className="hover:text-indigo-600 transition">AI Tutor</a></li>
            <li><a href="#contact" className="hover:text-indigo-600 transition">Contact</a></li>
          </ul>

          <button className="px-4 py-2 bg-indigo-600 text-white rounded-md shadow-md hover:bg-indigo-700 transition">
```

```
            Login
          </button>

          {/* Mobile Hamburger */}
          <div
            className="md:hidden flex flex-col gap-1 cursor-pointer"
            onClick={() => setMenuOpen(!menuOpen)}
          >
            <span className="w-6 h-0.5 bg-gray-700"></span>
            <span className="w-6 h-0.5 bg-gray-700"></span>
            <span className="w-6 h-0.5 bg-gray-700"></span>
          </div>
        </div>

        {/* Mobile Menu */}
        {menuOpen && (
          <div className="flex flex-col items-center bg-white shadow-md py
              -6 space-y-4 md:hidden">
            <a href="#features" className="text-gray-700 hover:text-indigo
                -600">Features</a>
            <a href="#steps" className="text-gray-700 hover:text-indigo-600
                ">Steps</a>
            <a href="#psychotest" className="text-gray-700 hover:text-
                indigo-600">Career Test</a>
            <a href="#about" className="text-gray-700 hover:text-indigo-600
                ">About</a>
            <a href="#testimonials" className="text-gray-700 hover:text-
                indigo-600">Testimonials</a>
            <a href="#pricing" className="text-gray-700 hover:text-indigo
                -600">Pricing</a>
            <a href="#contact" className="text-gray-700 hover:text-indigo
                -600">Contact</a>
          </div>
        )}
      </nav>

      {/* Hero Section */}
      <section id="hero" className="min-h-screen flex items-center bg-
          gradient-to-br from-gray-50 to-indigo-50 pt-24">
        <div className="max-w-7xl mx-auto grid md:grid-cols-2 gap-12 items-
            center px-6">
          <div>
            <h1 className="text-4xl md:text-5xl font-bold text-gray-900
                leading-tight mb-6">
              AI-Powered Career Roadmaps
            </h1>
            <p className="text-lg text-gray-600 mb-8">
              Plan your career path with personalized, step-by-step
```

```
                guidance.
          </p>
          <div className="flex gap-4 flex-wrap mb-10">
            <a href="#psychotest">
              <button className="px-6 py-3 bg-indigo-600 text-white rounded-lg shadow-md hover:bg-indigo-700 transition">
                Take the Career Test
              </button>
            </a>
            <button className="px-6 py-3 border-2 border-indigo-600 text-indigo-600 rounded-lg hover:bg-indigo-600 hover:text-white transition">
              Learn More
            </button>
          </div>
        </div>

        <div className="flex justify-center">
          <div className="w-72 h-72 bg-indigo-600 rounded-2xl flex items-center justify-center shadow-xl animate-bounce">
            <FaCompass className="text-white text-6xl" />
          </div>
        </div>
      </div>
    </section>

    {/* Career Demand Section (Pie Chart) */}
    <section id="career-demand" className="py-20 bg-white">
      <div className="max-w-7xl mx-auto px-6 grid md:grid-cols-2 gap-12 items-center">
        <div className="w-full bg-white rounded-xl shadow-lg p-6">
          <h3 className="text-xl font-semibold text-gray-800 mb-4 text-center">
            Career Demand by Domain
          </h3>
          <ResponsiveContainer width="100%" height={300}>
            <PieChart>
              <Pie
                data={chartData}
                cx="50%"
                cy="50%"
                outerRadius={100}
                dataKey="value"
                label
              >
                {chartData.map((entry, index) => (
                  <Cell key={index} fill={COLORS[index % COLORS.length]} />
```

```
                ))}
              </Pie>
              <Tooltip />
            </PieChart>
          </ResponsiveContainer>

          {/* Legend with icons + descriptions */}
          <div className="mt-6 grid grid-cols-2 gap-4 text-sm">
            <LegendItem icon={<FaCode />} color={COLORS[0]} title="Web Dev" desc="25% demand    Full-stack & frontend roles" />
            <LegendItem icon={<FaDatabase />} color={COLORS[1]} title="Data Science" desc="20% demand    AI, ML & Analytics" />
            <LegendItem icon={<FaBriefcase />} color={COLORS[2]} title="Business" desc="15% demand    Analysts & Consultants" />
            <LegendItem icon={<FaServer />} color={COLORS[3]} title="DevOps" desc="10% demand    Cloud & automation experts" />
            <LegendItem icon={<FaRobot />} color={COLORS[4]} title="AI" desc="20% demand    ML, NLP & AI Engineers" />
            <LegendItem icon={<FaLaptopCode />} color={COLORS[5]} title="Programming" desc="10% demand    Core language experts" />
          </div>
        </div>
        <div>
          <h2 className="text-3xl font-bold text-gray-900 mb-4">Why Career Demand Matters?</h2>
          <p className="text-gray-600 mb-4">
            Understanding market demand helps you choose a career with strong growth and stability.
            AI-powered insights ensure you stay ahead in competitive fields.
          </p>
          <p className="text-gray-600">
            This analysis reflects the skills most valued in the industry today. By aligning your learning path
            with these trends, you maximize opportunities for success.
          </p>
        </div>
      </div>
    </section>

    {/* Features Section */}
    <section id="features" className="py-20 bg-gray-50">
      <div className="max-w-7xl mx-auto px-6">
        <h2 className="text-3xl md:text-4xl font-bold text-gray-900 text-center mb-12">
          Features
        </h2>
        <div className="grid gap-10 sm:grid-cols-2 md:grid-cols-3">
```

```
            {featuresData.map((feature, idx) => (
              <Feature
                key={idx}
                icon={
                  feature.title === "AI Insights" ? (
                    <FaBrain />
                  ) : feature.title === "Career Roadmap" ? (
                    <FaRoad />
                  ) : (
                    <FaChartLine />
                  )
                }
                title={feature.title}
                description={feature.description}
                onClick={() => {
                  setSelectedFeature(feature);
                  setIsModalOpen(true);
                }}
              />
            ))}
          </div>
        </div>
      </section>

      {/* Steps Section */}
      <section id="steps" className="py-20 bg-white">
        <div className="max-w-7xl mx-auto px-6">
          <h2 className="text-3xl md:text-4xl font-bold text-gray-900 text-center mb-12">
            How It Works
          </h2>

          <div className="flex justify-center mb-8">
            <button
              onClick={() => setIsStepsOpen(true)}
              className="px-6 py-3 bg-indigo-600 text-white rounded-lg shadow-md hover:bg-indigo-700 transition"
            >
              Explore Steps
            </button>
          </div>

          <div className="grid gap-10 md:grid-cols-3">
            <Step number="1" title="Tell Us Your Interests" description="Fill out a simple form about your passions, skills, and goals." />
            <Step number="2" title="Choose Your Path" description="Get AI-powered recommendations for your career journey." />
```

```
            <Step number="3" title="Receive Your Roadmap" description="Access a step-by-step career plan tailored just for you." />
          </div>
        </div>
      </section>

      {isStepsOpen && (
        <StepsModal isOpen={isStepsOpen} onClose={() => setIsStepsOpen(false)} />
      )}

      {/* Psychometric Test Section */}
      <section id="psychotest" className="py-20 bg-gray-50">
        <div className="max-w-7xl mx-auto px-6">
          <h2 className="text-3xl md:text-4xl font-bold text-gray-900 text-center mb-12">
            Take Your Career Test
          </h2>
          <PsychometricTest />
        </div>
      </section>

      {/* About Section */}
      <section id="about" className="py-20 bg-white">
        <div className="max-w-7xl mx-auto px-6 grid md:grid-cols-2 gap-12 items-center">
          <div>
            <h2 className="text-3xl md:text-4xl font-bold text-gray-900 mb-6">
              About CareerMap
            </h2>
            <p className="text-lg text-gray-600 mb-6">
              We are dedicated to helping students and professionals discover
              their true potential by offering AI-driven career roadmaps. Our
              mission is to empower individuals with the right knowledge and
              tools to achieve success.
            </p>
            <p className="text-lg text-gray-600 mb-8">
              With smart insights and step-by-step planning, CareerMap makes
              career guidance simple, modern, and effective.
            </p>
            <div className="grid grid-cols-3 gap-6">
              <StatCard number="10K+" label="Users Guided" />
              <StatCard number="500+" label="Career Paths" />
```

```
              <StatCard number="95%" label="Satisfaction Rate" />
            </div>
          </div>
          <div className="flex justify-center">
            <div className="w-80 h-80 bg-indigo-600 rounded-2xl flex items-center justify-center shadow-lg">
              <FaUserGraduate className="text-white text-7xl" />
            </div>
          </div>
        </div>
      </section>

      {/* Testimonials Section */}
      <section id="testimonials" className="py-20 bg-gray-50">
        <div className="max-w-7xl mx-auto px-6 text-center">
          <h2 className="text-3xl md:text-4xl font-bold text-gray-900 mb-12">
            What Our Users Say
          </h2>
          <div className="grid gap-8 sm:grid-cols-2 lg:grid-cols-3">
            <Testimonial img="https://i.pravatar.cc/100?img=1" name="Priya Sharma" role="Engineering Student" text="CareerMap helped me figure out the right career path when I was lost. The roadmap made everything clear!" />
            <Testimonial img="https://i.pravatar.cc/100?img=2" name="Rahul Verma" role="MBA Graduate" text="The AI-powered insights were spot on! It gave me confidence to pursue my chosen career." />
            <Testimonial img="https://i.pravatar.cc/100?img=3" name="Anjali Patel" role="Software Developer" text="I loved the clean design and easy-to-follow roadmap. Highly recommend CareerMap!" />
          </div>
        </div>
      </section>

      {/* Contact Section */}
      <section id="contact" className="py-20 bg-white">
        <div className="max-w-7xl mx-auto px-6 grid md:grid-cols-2 gap-12 items-center">
          <div>
            <h2 className="text-3xl md:text-4xl font-bold text-gray-900 mb-6">
              Get in Touch
            </h2>
            <p className="text-lg text-gray-600 mb-8">
              Have questions or need guidance?  W e re  here to help you take the
```

```
              next step in your career journey.
            </p>
            <div className="space-y-4">
              <ContactInfo icon={<FaEnvelope />} text="support@careermap.com" />
              <ContactInfo icon={<FaPhone />} text="+91 98765 43210" />
              <ContactInfo icon={<FaMapMarkerAlt />} text="Mumbai, India" />
            </div>
          </div>
          <div className="bg-white rounded-xl shadow-md p-8">
            <form className="space-y-6">
              <FormInput label="Name" type="text" placeholder="Your Name" />
              <FormInput label="Email" type="email" placeholder="Your Email" />
              <FormTextarea label="Message" placeholder="Write your message..." />
              <button type="submit" className="w-full px-6 py-3 bg-indigo-600 text-white font-semibold rounded-lg shadow-md hover:bg-indigo-700 transition">
                Send Message
              </button>
            </form>
          </div>
        </div>
      </section>

      {/* Footer */}
      <footer className="bg-gray-900 text-gray-400 py-10 mt-20">
        <div className="max-w-7xl mx-auto px-6 flex flex-col md:flex-row justify-between items-center gap-8">
          <div className="flex items-center text-indigo-500 font-bold text-xl">
            <FaMap className="mr-2" /> CareerMap
          </div>
          <div className="flex gap-6 flex-wrap justify-center">
            <a href="#features" className="hover:text-indigo-400 transition">Features</a>
            <a href="#steps" className="hover:text-indigo-400 transition">Steps</a>
            <a href="#about" className="hover:text-indigo-400 transition">About</a>
            <a href="#pricing" className="hover:text-indigo-400 transition">Pricing</a>
            <a href="#contact" className="hover:text-indigo-400 transition">Contact</a>
          </div>
```

```
          <div className="flex gap-4">
            <a href="#" className="w-10 h-10 bg-gray-800 rounded-full flex
                items-center justify-center hover:bg-indigo-600 transition"
                ><FaTwitter /></a>
            <a href="#" className="w-10 h-10 bg-gray-800 rounded-full flex
                items-center justify-center hover:bg-indigo-600 transition"
                ><FaLinkedinIn /></a>
            <a href="#" className="w-10 h-10 bg-gray-800 rounded-full flex
                items-center justify-center hover:bg-indigo-600 transition"
                ><FaFacebookF /></a>
          </div>
        </div>
        <div className="text-center text-gray-500 text-sm mt-8">
             {new Date().getFullYear()} CareerMap. All rights reserved.
        </div>
      </footer>

      {/* Feature Modal */}
      <FeatureModal
        isOpen={isModalOpen}
        onClose={() => setIsModalOpen(false)}
        title={selectedFeature?.title}
        description={selectedFeature?.description}
        roadmap={selectedFeature?.roadmap}
      />
    </div>
  );
}

/* ------------------ Reusable Components ------------------ */

function Feature({ icon, title, description, onClick }) {
  return (
    <div
      onClick={onClick}
      className="cursor-pointer bg-white rounded-xl shadow-md p-6 hover:
          shadow-xl transition flex flex-col items-center text-center"
    >
      <div className="text-indigo-600 text-4xl mb-4">{icon}</div>
      <h3 className="text-xl font-semibold text-gray-800 mb-2">{title}</h3>
      <p className="text-gray-600">{description}</p>
    </div>
  );
}

function Step({ number, title, description }) {
  return (
    <div className="bg-gray-50 p-6 rounded-xl shadow-md hover:shadow-lg
```

```
        transition">
      <div className="w-10 h-10 flex items-center justify-center rounded-
        full bg-indigo-600 text-white text-lg font-bold mb-4">
        {number}
      </div>
      <h3 className="text-lg font-semibold text-gray-800 mb-2">{title}</h3>
      <p className="text-gray-600">{description}</p>
    </div>
  );
}

function StatCard({ number, label }) {
  return (
    <div className="bg-gray-50 p-6 rounded-lg shadow-md text-center">
      <h3 className="text-2xl font-bold text-indigo-600">{number}</h3>
      <p className="text-gray-600">{label}</p>
    </div>
  );
}

function Testimonial({ img, name, role, text }) {
  return (
    <div className="bg-white rounded-xl shadow-md p-6 flex flex-col items-
      center text-center">
      <img src={img} alt={name} className="w-16 h-16 rounded-full mb-4" />
      <h3 className="text-lg font-semibold text-gray-800">{name}</h3>
      <p className="text-sm text-indigo-600 mb-2">{role}</p>
      <p className="text-gray-600">{text}</p>
    </div>
  );
}

function ContactInfo({ icon, text }) {
  return (
    <div className="flex items-center gap-4 text-gray-700">
      <div className="text-indigo-600">{icon}</div>
      <span>{text}</span>
    </div>
  );
}

function FormInput({ label, type, placeholder }) {
  return (
    <div>
      <label className="block text-gray-700 font-medium mb-2">{label}</
        label>
      <input
        type={type}
```

```
        placeholder={placeholder}
        className="w-full px-4 py-2 border rounded-lg focus:ring-2 focus:ring-indigo-500 outline-none"
      />
    </div>
  );
}

function FormTextarea({ label, placeholder }) {
  return (
    <div>
      <label className="block text-gray-700 font-medium mb-2">{label}</label>
      <textarea
        placeholder={placeholder}
        rows="4"
        className="w-full px-4 py-2 border rounded-lg focus:ring-2 focus:ring-indigo-500 outline-none"
      ></textarea>
    </div>
  );
}

function LegendItem({ icon, color, title, desc }) {
  return (
    <div className="flex items-start gap-3">
      <div
        className="w-5 h-5 flex items-center justify-center rounded-full"
        style={{ backgroundColor: color }}
      >
        <div className="text-white text-xs">{icon}</div>
      </div>
      <div>
        <h4 className="font-medium text-gray-800">{title}</h4>
        <p className="text-gray-600 text-xs">{desc}</p>
      </div>
    </div>
  );
}
```